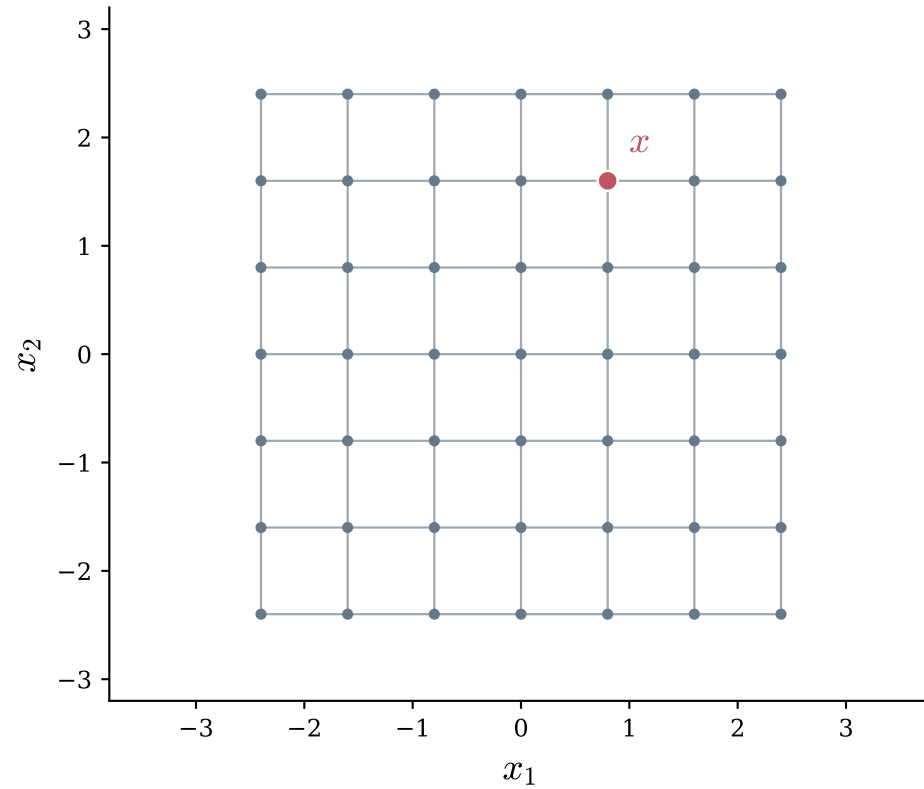


A parameter direction becomes a velocity field on the transported grid

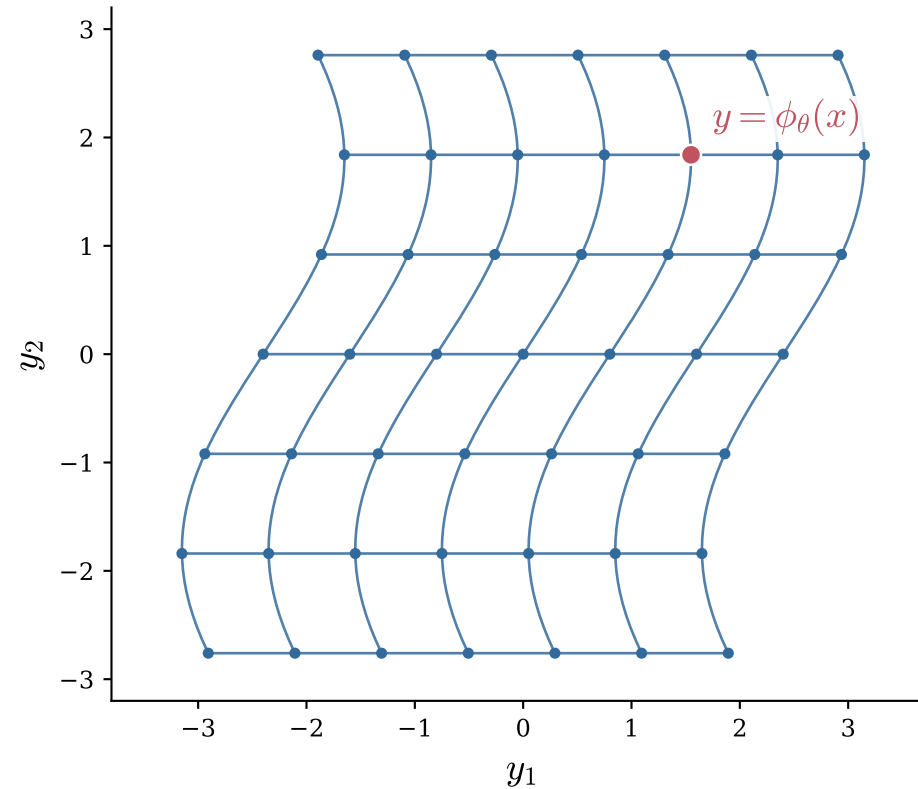
$$\phi_\theta(x_1, x_2) = (x_1 + \theta_1 \sin x_2, \theta_2 x_2), \quad \theta = (0.75, 1.15), \quad d\theta = (0.85, -0.45)$$

$$\theta(t) = \theta + t d\theta, \quad \theta'(t) = d\theta, \quad \varepsilon = 0.45$$

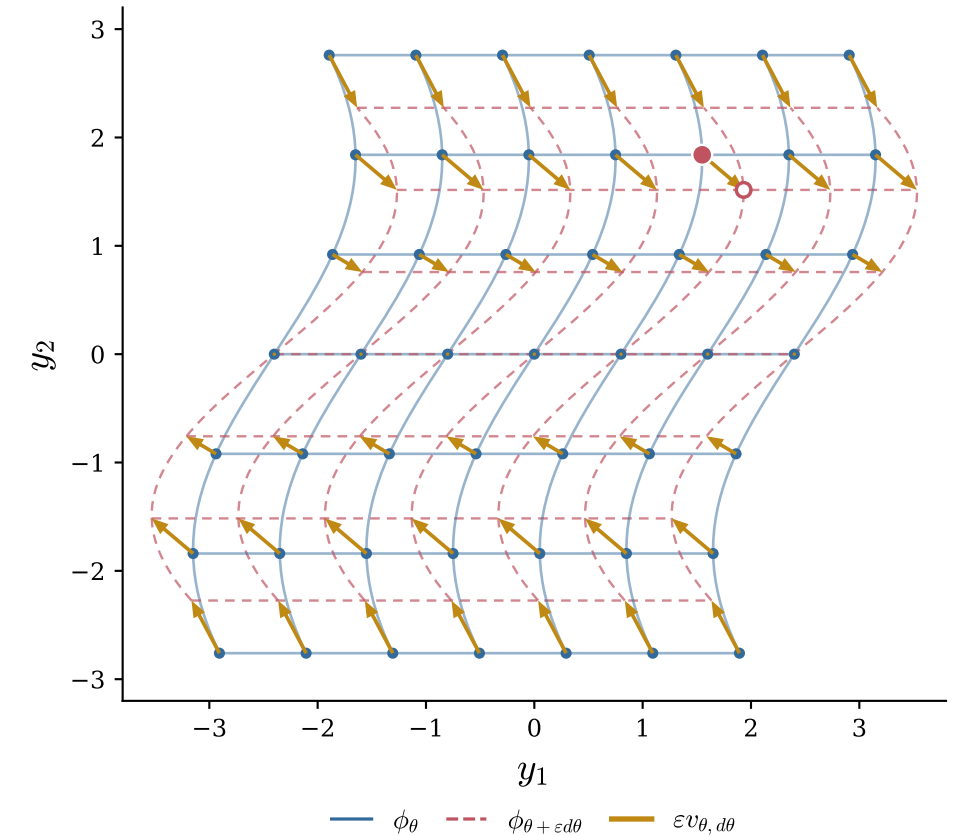
(a) Reference grid



(b) Current transported grid



(c) Velocity and the next grid



$$v_{\theta, d\theta}(\phi_\theta(x)) = d_\theta \phi_\theta(d\theta)(x) = (d\theta_1 \sin x_2, d\theta_2 x_2)$$

$$v_{\theta, d\theta}(y) = \left(d\theta_1 \sin\left(\frac{y_2}{\theta_2}\right), d\theta_2 \frac{y_2}{\theta_2} \right)$$

Arrows start at current positions and show the velocity multiplied by ε . The red dot tracks one reference node.

Here the map is affine in the parameters: every arrow tip lies exactly on the corresponding next grid node.